Contraction Theory Based Control Under Input Saturation–LMI Approach ⋆

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Abstract: Besides the traditional Lyapunov stability theory, contraction theory offers a interesting perspective on the stability of nonlinear systems. It allows us analyze the system stability by examining the convergence of trajectories. The recent development of the contraction theory based control has introduced computational efficient methods for obtaining contraction metrics using linear matrix inequalities (LMIs). In this paper, the contraction theory based control strategy is extended to account for input saturation by incorporating saturation constraints into the LMI framework for contraction metric calculation. The proposed method calculates the contraction metrics while ensuring the satisfaction of input saturation constraints and uses as a feedback mechanism to adjust the control inputs. The proposed method is demonstrated its effectiveness through numerical simulations via Lorenz system.

Keywords: Contraction theory, input saturation, linear matrix inequalities (LMIs)

NOTATION

In this paper, we use the following notation:

- $:=$ denotes *defined as*.
- $(\cdot^\top)$ denotes the transpose of a matrix or a vector.
- $\boldsymbol{x} := [x_i]_{i\in\{1,\cdots,n\}} \in \mathbb{R}^n$ denotes the state vector.
- $\boldsymbol{A} := [a_{ij}]_{i,j\in\{1,\cdots,n\}} \in \mathbb{R}^{n\times n}$ denotes a matrix.
- $\lambda_i(\boldsymbol{A})$, $i \in \{\max, \min\}$ denotes the maximum and minimum singular value of $\boldsymbol{A}$, respectively.
- $\boldsymbol{I}_n$ denotes the identity matrix of size n and $\mathbf{0}_{n\times m}$ denotes the zero matrix of size $n \times m$.
- sym denotes the symmetric part of a matrix, i.e., $\text{sym}(\boldsymbol{A}) := \frac{1}{2}(\boldsymbol{A} + \boldsymbol{A}^\top)$ (see Tsukamoto et al. (2021b)).

1. INTRODUCTION

1.1 Motivation

- Contraction theory introduction
- Limitation 1 (computational cost)
- Recent development (LMI)
- Limitation 2 (input saturation)
  - · use penalty term to prevent excessive control input
- saturation handling in LMI

To the best of our knowledge, there has been no previous work that considers the input saturation in LMI frameworks for contraction metrics. Hence, the main objective of this paper is to extend the contraction theory based control strategy to account for input saturation simultaneously in the LMI framework.

1.2 Contributions

The main contributions of this paper are

- The contraction metric is obtained by using linear matrix inequalities (LMI) which simplifies the calculation of the contraction metric;
- Saturation is incorporated into the LMI framework for contraction metric calculation;
- ...; and
- The effectiveness and the feasibility of the proposed method is validated through numerical simulations via Lorenz system.

2. PRELIMINARIES

Before presenting the proposed method, we first review the contraction theory and related theorems.

2.1 Contraction Theory

Consider that we have the following deterministic nonlinear system:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, t). \tag{1}$$

The contraction theory starts with the definition of *differential dynamics* of the system (1) using *differential displacement* $\delta\boldsymbol{x}$ (Kirk, 2004, Chap 4), which is defined as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\delta\boldsymbol{x} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\delta\boldsymbol{x}. \tag{2}$$

⋆ Sponsor and financial support acknowledgment goes here. Paper titles should be written in uppercase and lowercase letters, not all uppercase.

The differential displacement $\delta\boldsymbol{x}$ is the infinitesimal displacement at *fixed time*.

The contraction property of the system (1) can be analyzed using the Lyapunov function $V := V(\boldsymbol{x},t) = \frac{1}{2}\delta\boldsymbol{x}^\top \boldsymbol{M}\delta\boldsymbol{x}$, where $\boldsymbol{M} := \boldsymbol{M}(\boldsymbol{x},\boldsymbol{x}_d,t) \in \mathbb{R}^{n\times n}$ the *contraction metric*. The contraction metric $\boldsymbol{M}$ is symmetric positive definite matrix and satisfies $\overline{m}\boldsymbol{I}_n \succeq \boldsymbol{M} \succeq \underline{m}\boldsymbol{I}_n$ with $\overline{m},\underline{m} \in \mathbb{R}_{>0}$. For the system (1) to be contracting, the following condition must hold:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + 2\,\mathrm{sym}(\boldsymbol{M}\tfrac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}) \leq -2\alpha\boldsymbol{M}, \tag{3}$$

which leads to $\frac{\mathrm{d}}{\mathrm{d}t}V \leq -\alpha V$ for some $\alpha \in \mathbb{R}_{>0}$. In other words, all solutions of the system (1) are contracting to a unique solution with an exponential rate α.

2.2 Partial Contraction Theory

On the other hand, the partial contraction theory investigates the contraction of the system using a virtual system. Following theorem details the partial contraction theory.

Theorem 1. (see Wang and Slotine (2004)). Consider a nonlinear system of the form $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x},\boldsymbol{x},t)$ and the virtual system $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{q} = \boldsymbol{f}(\boldsymbol{q},\boldsymbol{x},t)$ are contracting with respect to $\boldsymbol{q}$. If a particular solution of virtual system verifies a smooth specific property, then all solutions of the original system verifies this property exponentially.

Proof. Assume that the possible solutions of the virtual system $\boldsymbol{q}$ are $\boldsymbol{\xi}_1(t)$ and $\boldsymbol{\xi}_2(t)$ which are the trajectories of the nonlinear system $\boldsymbol{x}$. ...

2.3 Perturbed Contracting System

The following theorem presents the contraction property of the perturbed system by adding a bounded perturbation $\boldsymbol{d}(\boldsymbol{x},t)$ to the system (1):

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x},t) + \boldsymbol{d}(\boldsymbol{x},t). \tag{4}$$

Theorem 2. (Eq. 15 Lohmiller and Slotine (1998)). If the system (1) is contracting with the contraction metric $\boldsymbol{M}$ which satisfies the condition (3), the path integral of the virtual system ... [MS: ADD!!]

$$\|\boldsymbol{\xi}_1(t) - \boldsymbol{\xi}_2(t)\| \leq \tfrac{V_l(0)}{\sqrt{\underline{m}}}\exp(-\alpha t) + \tfrac{\overline{d}}{\alpha}\sqrt{\tfrac{\overline{m}}{\underline{m}}}(1-\exp(-\alpha t)), \tag{5}$$

[MS: ADD!!]

Proof. See Theorem 2.4 in Tsukamoto et al. (2021c).

[MS: Any more comments, if needed]

3. PROBLEM FORMULATION

Let us consider the following nonlinear system as follows:

$$\begin{aligned}\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} =& \boldsymbol{f}(\boldsymbol{x},t) + \boldsymbol{B}(\boldsymbol{x},t)\,\mathrm{sat}(\boldsymbol{u}) + \boldsymbol{d}(t)\\ =& \boldsymbol{f}(\boldsymbol{x},t) + \boldsymbol{B}(\boldsymbol{x},t)\boldsymbol{u} + \boldsymbol{\phi}(\boldsymbol{u}) + \boldsymbol{d}(t),\end{aligned} \tag{6}$$

where $\boldsymbol{x} \in \mathbb{R}^n$ denotes the state variables, $\boldsymbol{u} \in \mathbb{R}^n$ denotes the control inputs, and $\boldsymbol{d}(t) \in \mathbb{R}^n$ denotes the external disturbances, and $\boldsymbol{f} := \boldsymbol{f}(\boldsymbol{x},t)$ and $\boldsymbol{B} := \boldsymbol{B}(\boldsymbol{x},t)$ denote system functions. The input saturation function is denoted by $\mathrm{sat}(\cdot)$ and is defined as

$$\mathrm{sat}(\boldsymbol{u}) = \begin{cases}\boldsymbol{u}, & \text{if } \|\boldsymbol{u}\| \leq \overline{u}\\ \frac{\boldsymbol{u}}{\|\boldsymbol{u}\|}, & \text{otherwise},\end{cases} \tag{7}$$

where $\overline{u} \in \mathbb{R}_{>0}$ denotes the saturation threshold. The difference between the actual control input $\boldsymbol{u}$ and the saturated control input is denoted by $\boldsymbol{\phi}(\boldsymbol{u}) = \mathrm{sat}(\boldsymbol{u}) - \boldsymbol{u}$.

In the practical perspective, we consider the following assumption:

Assumption 3. The control input matrix $\boldsymbol{B}$ is sign-definite, so that the inverse of $\boldsymbol{B}$ exists.

Motivated by the use of the partial contraction theory in Tsukamoto et al. (2021a), [MS: and more...], we assume that the desired trajectory (including desired unsaturated control input, i.e., $\|\boldsymbol{u}_d\| \leq \overline{u}$) of the system is given without disturbances, i.e., $d = 0$, as follows:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}_d = \boldsymbol{f}_d + \boldsymbol{B}_d\boldsymbol{u}_d, \tag{8}$$

where $\boldsymbol{x}_d \in \mathbb{R}^n$ is the desired state vector and $\boldsymbol{u}_d \in \mathbb{R}^n$ is the desired control input vector, and $\boldsymbol{f}_d := \boldsymbol{f}(\boldsymbol{x}_d,t)$ and $\boldsymbol{B}_d := \boldsymbol{B}(\boldsymbol{x}_d,t)$.

Finally, the control objective is to design a control input $\boldsymbol{u}$ such that the system trajectories of the system (8) and the desired system (8) converges, under the existences of the disturbances $\boldsymbol{d}$ and input saturation $\mathrm{sat}(\cdot)$.

4. MAIN RESULTS

Following the control design in Tsukamoto et al. (2021a), state-dependent coefficient (SDC) formulation Dani et al. (2015) is used to represent the error dynamics [MS: more details]. The SDC $\mathbb{A} := \mathbb{A}(\boldsymbol{x},\boldsymbol{x}_d)$ is defined as follows:

$$\mathbb{A} := \int_0^1 \left[\tfrac{\partial \overline{\boldsymbol{f}}}{\partial \boldsymbol{x}}\right](c\boldsymbol{x} + (1-c)\boldsymbol{x}_d)\,\mathrm{d}c, \tag{9}$$

where $\overline{\boldsymbol{f}} := \overline{\boldsymbol{f}}(\boldsymbol{x},\boldsymbol{x}_d) = \boldsymbol{f} - \boldsymbol{f}_d + \boldsymbol{B}\boldsymbol{u}_d - \boldsymbol{B}_d\boldsymbol{u}_d$ and $\boldsymbol{B}_d := \boldsymbol{B}(\boldsymbol{x}_d)$, and c is a dummy variable for the integration. The existence... blah blah can be found in Dani et al. (2015), [MS: and more...].

Then, the error dynamics can be expressed as:

$$\begin{aligned}\tfrac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{x}-\boldsymbol{x}_d) =& \boldsymbol{f} - \boldsymbol{f}_d + \boldsymbol{B}\boldsymbol{u} - \boldsymbol{B}_d\boldsymbol{u}_d + \boldsymbol{\phi}(\boldsymbol{u}) + \boldsymbol{d}\\ =& \underbrace{\boldsymbol{f} - \boldsymbol{f}_d + (\boldsymbol{B} - \boldsymbol{B}_d)\boldsymbol{u}_d}_{=\mathbb{A}(\boldsymbol{x},\boldsymbol{x}_d)(\boldsymbol{x}-\boldsymbol{x}_d)}\\ &- \boldsymbol{B}(\boldsymbol{u}-\boldsymbol{u}_d) + \boldsymbol{\phi}(\boldsymbol{u}) + \boldsymbol{d}\\ =& \mathbb{A}(\boldsymbol{x}-\boldsymbol{x}_d) - \boldsymbol{B}(\boldsymbol{u}-\boldsymbol{u}_d) + \boldsymbol{\phi}(\boldsymbol{u}) + \boldsymbol{d}\\ =& \left(\mathbb{A} - \boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top\boldsymbol{M}\right)(\boldsymbol{x}-\boldsymbol{x}_d) + \boldsymbol{\phi}(\boldsymbol{u}) + \boldsymbol{d},\end{aligned} \tag{10}$$

with the control input $\boldsymbol{u}$ defined as:

$$\boldsymbol{u} = \boldsymbol{u}_d - \boldsymbol{R}^{-1}\boldsymbol{B}^\top\boldsymbol{M}(\boldsymbol{x}-\boldsymbol{x}_d) \tag{11}$$

where $\boldsymbol{R} \succ 0$ is a positive definite matrix and $\boldsymbol{M} := \boldsymbol{M}(\boldsymbol{x},\boldsymbol{x_d},t) \succ 0$ is the contraction metric defined in Section 2.1.

4.1 Contraction Metric Calculation

Contraction Condition Constraint Invoking the partial contraction theory reviewed in Section 2.1, we can define the virtual system as follows:

$$\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{q} = \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}_d + \left(\mathbb{A} - \boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top\boldsymbol{M}\right)(\boldsymbol{q}-\boldsymbol{x}_d) + d_\mu, \tag{12}$$

where $\boldsymbol{q}$ is the virtual state vector and $d_\mu := d_\mu(d,\mu), \forall\mu \in [0,1]$ is defined as $d_\mu = d_\mu(d,0)$ and $d_\mu = d_\mu(d,1)$. The differential dynamics is then defined as:

$$\frac{\mathrm{d}}{\mathrm{d}t}\delta \boldsymbol{q} = \left(\mathbb{A} - \boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top \boldsymbol{M}\right)\delta \boldsymbol{q}. \tag{13}$$

Then according to Theorem 2 and the partial contraction theory, the system (12) and the system (8) will converge to the desired trajectory $\boldsymbol{x}_d$ with steady state error:

$$\boldsymbol{e}_{\mathrm{ss}} = ..?, \tag{14}$$

with the following conditions:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M} + 2\,\mathrm{sym}(\boldsymbol{M}\mathbb{A}) + 2\boldsymbol{M}\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{B}^\top\boldsymbol{M} \leq -2\alpha\boldsymbol{M}. \tag{15}$$

In [MS: []], Tsukamoto et al.proposed a method to formulate the LMI problem for the contraction metric as follows:

$$\begin{aligned} &\min_{\boldsymbol{W},X,\mu} \; X + \lambda\mu \\ &\text{s.t. } (15) \end{aligned} \tag{16}$$

However, even with the proposed method, the input saturation can still occur by excessive contraction metric leading to potential instability. Therefore, we need to incorporate input constraints into the LMI formulation.

Input Saturation Constraint To address the input saturation function (7), the saturation constraint $\boldsymbol{u}^\top\boldsymbol{u} - \overline{u}^2 \leq 0$ is defined as:

$$\begin{bmatrix} \overline{u}^2 & \left(\boldsymbol{u}_d - \boldsymbol{R}^{-1}\boldsymbol{B}^\top\boldsymbol{M}(\boldsymbol{x}-\boldsymbol{x}_d)\right)^\top \\ \star & \boldsymbol{I} \end{bmatrix} \succeq 0. \tag{17}$$

By multiplying both sides by $[\mathrm{diag}(\,\boldsymbol{I}, \boldsymbol{W}(\boldsymbol{R}^{-1}\boldsymbol{B}^\top)^{-1}\,)]$, we have LMIs as follows:

$$\begin{bmatrix} \overline{u}^2 & \left(\boldsymbol{W}(\boldsymbol{R}^{-1}\boldsymbol{B}^\top)^{-1}\boldsymbol{u}_d - (\boldsymbol{x}-\boldsymbol{x}_d)\right)^\top \\ \star & \mathbb{X} \end{bmatrix} \succeq 0, \tag{18}$$

where $\mathbb{X} := \boldsymbol{W}(\boldsymbol{R}^{-1}\boldsymbol{B}^\top)^{-1}{(\boldsymbol{R}^{-1}\boldsymbol{B}^\top)^{-1}}^\top\boldsymbol{W}$ which satisfies

$$\begin{bmatrix} \mathbb{X} & \boldsymbol{W}((\boldsymbol{R}^{-1}\boldsymbol{B}^\top)^{-1})^\top \\ \star & \boldsymbol{I} \end{bmatrix} \succeq 0. \tag{19}$$

Finally, the contraction metric can be obtained by solving the following LMI problem:

$$\begin{aligned} &\min_{\boldsymbol{W},X,\mu} \; X + \mathrm{trace}(\mathbb{X}) \\ &\text{s.t. } (15),\ (18) \text{ and } (19) \end{aligned} \tag{20}$$

4.2 Numerical Methods For LMI

- Discretize $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{M}$
- LMI solver
- Tuning methods

5. VALIDATIONS

We employed the Lorenz system [MS: REF] to validate the proposed method which is represented as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{=:\boldsymbol{\xi}} = \underbrace{\begin{pmatrix} \sigma(y-x) \\ x(\rho-z)-y \\ xy-\beta z \end{pmatrix}}_{=:\boldsymbol{f}(\boldsymbol{\xi})} + \underbrace{\begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix}}_{=:\boldsymbol{u}} + \underbrace{\begin{pmatrix} d_x \\ d_y \\ d_z \end{pmatrix}}_{=:\boldsymbol{d}}, \tag{21}$$

where $\sigma = 10$, $\rho = 28$, and $\beta = \frac{8}{3}$ are system parameters.

5.1 Scenario 1: Contraction Property

5.2 Scenario 2: Saturation Handling

6. CONCLUSION

In this study, ...

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Place acknowledgments here.

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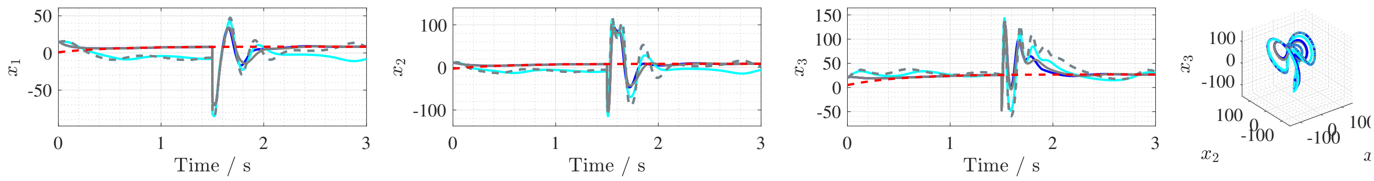
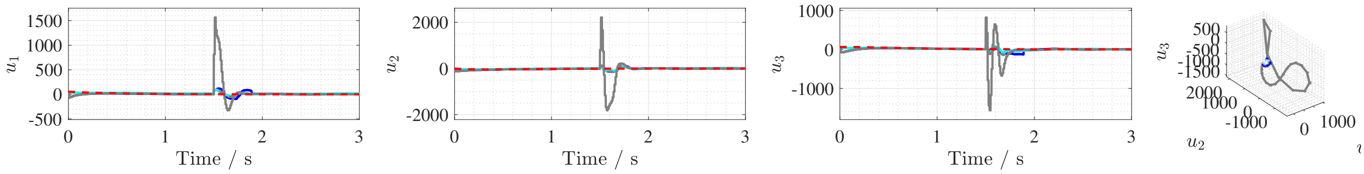


(a) Tracking performance of 1st joint angle q_1.



(b) Tracking performance of 2nd joint angle q_2.

Fig. 1. Real-time implementation results of (C_1) [▬], (C_2) [▬], (C_3) [▬], (C_4) [▬], and desired trajectory x_d [- - -].